



The influence of electrical charge on plasmodesma conductivity

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Most plant tissues are symplasms in which cells are connected by plasmodesmata, membrane-lined cytoplasmic bridges that enable diffusive and/or advective cell-to-cell movements of components of the cytosol. Current models assume that hydrodynamic radius alone governs the mobility of molecules through plasmodesmata. In contrast, physical theory predicts that nm-sized pores with electrically charged walls are selective for counterions. Narrow plasmodesmal pores lined by membranes—which carry negative surface charges—therefore should be permselective for cations. Quantifying cell-to-cell movements of fluorophores varying in molecular mass and electrical charge in cell types with differently sized plasmodesmata, we confirmed the applicability of physical theory to plasmodesmata. Surprisingly, narrow plasmodesmata were permselective for anions rather than cations, suggesting a major flaw in our current understanding of plasmodesma structure. We hypothesize that structural proteins known to exist in plasmodesmata may establish the cationic electrostatic environment required to explain our findings.

plasmodesma | nano channel | cell cell movement

In multicellular organisms, cellular activities have to be coordinated to accomplish developmentally and physiologically meaningful results (1). In many cases, neighboring cells communicate via direct physical connections that establish cytoplasmic continuity (2). Plant cells are encased in comparatively rigid cell walls, but still remain linked through cell wall pores that are traversed by cytoplasmic bridges (3–7). These plasmodesmata vary considerably in size, but often are nanoscopic with diameters of 50 nm or less (8). Not only plasma membrane and cytosol are continuous through plasmodesmata but also the endoplasmic reticulum (ER), which forms the so-called desmotubule in the center of the plasmodesmal pore (9). The open space between the peripheral plasma membrane and the central desmotubule membrane provides the route for symplasmic cell-to-cell transport. However, the small cross-sectional area of this cytosolic sleeve, which may be further restricted by structural proteins that connect desmotubule and plasma membrane (10, 11), creates steric constraints on particle movement (12). Therefore, a plasmodesma's size exclusion limit—the mass of the largest molecule capable of crossing from cell to cell through the cytosolic sleeve—is considered an essential functional characteristic (13, 14).

Currently available electron-microscopy methods are insufficient for estimating size exclusion limits from ultrastructural data (8). For example, not only the relatively small fluorophore carboxyfluorescein (0.38 kDa, 0.61 nm hydrodynamic radius) (15) but also the much larger green fluorescent protein (GFP; 27 kDa, 2.82 nm hydrodynamic radius) (16) diffuse through plasmodesmata in postcytokinetic root cap cells that show no cytosolic sleeves in electron micrographs at all (17). Movements of fluorescent tracers in living plants provide physiologically more meaningful insights than ultrastructural data alone. According to numerous tracer studies, plasmodesmal size exclusion limits differ significantly between cell types and developmental stages, but generally are greater in young or embryonic tissues (18, 19). The symplasmic connectivity of young root epidermis cells decreases when the cells mature (20). In leaves, GFP moves freely from cell to cell in sink tissues (developing tissues that still import photosynthates) but is significantly hindered in movement in source tissues (mature tissues that export photosynthates) (21). Comparatively large size exclusion limits also were detected in specialized cell types functioning in long-distance transport (22), including pore-plasmodesma units at the interface between sieve elements and companion cells (~70 kDa) and funnel plasmodesmata connecting protophloem sieve elements and the phloem pole pericycle in roots (>110 kDa) (12, 23). The bulk of the living mass of post-seedling-stage plants consists of mature cells not specialized for long-distance transport. The majority of studies showed that plasmodesmal size exclusion limits are below 3 kDa in mature tissues, too low to permit passage of the smallest currently available genetically encoded, proteinaceous tracers (~10 kDa) (24). To study plasmodesma transport in these tissues, small fluorophores such as

Significance

For decades, it was believed that the hydrodynamic radius of a molecule is the defining factor for diffusion or convection through plasmodesmata between plant cells. Here, we show that positively charged molecules are excluded from narrow plasmodesmata and that diffusivity of neutral and negatively charged molecules can vary significantly. This has general implications for our understanding of plasmodesma structure as well as of various physiological processes including hormone transport, metabolite distribution (e.g., positive versus negative amino acids), long-distance signaling, and others.

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carboxyfluorescein or HPTS have been introduced into the cells (25, 26). For example, size exclusion limits between 0.7 and 3 kDa have been reported for *Tradescantia* stamen hair cells (27–29). Stamen hairs and epidermal trichomes are classical objects for quantitative analyses of plasmodesmal transport (27–31), mainly because complications due to three-dimensional tissue structure are avoided in these linear cell files (32).

The hydrophilic surfaces of cellular membranes, and in particular the cytosol-facing side of the plasma membrane, are negatively charged (33, 34). Negative surface charges enable electrostatic binding of specific proteins to membranes (35, 36) and, more generally, the formation of an electrical double layer, as cations in the solution next to the membrane are attracted, whereas anions are repelled (32, 37). If a membrane-coated pore such as a cytosolic sleeve has a diameter not much larger than the thickness of this layer, electrostatics will govern the behavior of charged particles within the pore; the pore will be permselective with greater permeability for cations than for similarly sized anions (38). The extent of permeability enhancement for cations is the product of an electrically increased solubility of the species within the pore space (equation 2 in Christensen et al. 2023) and an effective mass transfer coefficient for a similar neutral species (D/L in equation 3 in Christensen et al. 2023). These physical principles, which received little attention in plasmodesma studies so far, imply qualitatively different plasmodesmal permeabilities and size exclusion limits for anions, cations, and uncharged particles, depending on the ratio between cytosolic sleeve diameter and the thickness of the electrical double layer (Fig. 1).

The latter depends on the ionic strength of the cytosol within transporting plasmodesmata, which is hard to determine in vivo. Based on published membrane charge densities (39), 100 to 300 mM cytosolic ionic strength, and practical knowledge from artificial nano-porous materials (40), significant electrostatic effects will have to be expected only in cytosolic sleeves that are less than 3 to 4 nm wide. With a hydrodynamic diameter of almost 6 nm, GFP diffusion through plasmodesmata requires wider cytosolic sleeves for steric reasons alone, and the net charge of this protein (estimated to be -8 at a cytosolic pH, pH_{cyt} of 7.3) (41) is predicted to have little effect (compare Fig. 1 *A* and *E*), which aligns well with the limited empirical results available (42). On the other hand, the exclusion of GFP from many plasmodesmata indicates that the effective diameter of the space available for diffusion in the cytosolic sleeves of these plasmodesmata is under 6 nm. Thus it is a plausible hypothesis that membrane electrostatics may impact transport through narrow plasmodesmata in mature tissues (compare Fig. 1 *B–D* and *F–H*; please note: The movement of counter ions over membranes or via ion channels and the formation of possible streaming potentials have been omitted in the model for simplicity and clarity. The effects would be minimal at a concentration of charged fluorescent probes of 200 μ M [see below], which is orders of magnitude below the cytosolic concentrations of potential counter ions such as potassium or chloride). If corroborated, this could question current models of the transport of essential anionic compounds such as indole-3-acetic acid, the plant growth hormone auxin (43).

As electrical double layers form necessarily due to physical principles, two possibilities exist for narrow nanopores lined by charged membranes: With negative surfaces, anion movements will be significantly reduced or excluded (Fig. 1 *A–D*), while with positive surfaces, cation movements will be significantly reduced or excluded (Fig. 1 *E–H*). Current models assume that standard negatively charged membranes face the cytosolic sleeve, implying that anions should not be able to move through narrow plasmodesmata. While the potential significance for plasmodesmal function

of “electrostatic exclusion in the pore” occasionally has been recognized (44), the current view of nontargeted (i.e., diffusive or advective) plasmodesmal transport focuses almost exclusively on sterically defined size exclusion limits. In the influential work titled “Hydrodynamic radius alone governs the mobility of molecules through plasmodesmata”, plasmodesmal conductivity was assessed through cell-to-cell movements of the fluorescent dye fluorescein isothiocyanate (FITC) conjugated to various amino acids and short peptides (45). All conjugates tested were di- or trivalent anions at pH 8 (slightly above pH_{cyt}), but the potential significance of this fact was not evaluated. Subsequently, 27 FITC-conjugates were employed that mostly were anionic (up to pentavalent) while two were considered neutral (31). Here, the authors concluded that charge as well as molecular weight affected plasmodesmal diffusion, but that structural features of the diffusing molecules were most important (31). Very small fluorophores theoretically capable of permeating membranes can be used as symplasmic tracers nonetheless if they are charged and thus membrane-impermeant (25). For instance, carboxyfluorescein, which readily diffuses through most plasmodesmata at high rates (23, 30, 46), is anionic (mostly trivalent) at pH_{cyt} (47). Notably, this method should work with negatively as well as positively charged fluorophores, but intriguingly, 11 of the 13 small tracers that we found in the published record of plasmodesma studies are anionic at pH_{cyt} while none is cationic (*SI Appendix, Table S1*).

Taken together, the notion commonly held by plant cell biologists that the mobility of charged particles along nm-sized channels with charged borders (i.e., narrow cytosolic sleeves) depends on particle size alone is incompatible with physical theory and practical knowledge of analogous processes in artificial systems. One rather should expect qualitatively different transport characteristics in plasmodesmata that are too narrow to let fluorescent proteins pass, compared to wider ones. In particular, if current models of plasmodesmal ultrastructure were correct (scenarios B, C, and D in Fig. 1), narrow plasmodesmata should be permselective, favoring the movement of positively charged particles—however, almost all small tracers commonly used to evaluate symplasmic continuity are anionic. Here, we present quantitative in vivo-studies of plasmodesmal permeabilities for fluorophores varying in size and electrical charge that lead us to suggest a modified model of plasmodesma structure.

Results

Qualitative Analysis: Testing Predictions of Physical Theory in Various Tissues. Predictions concerning the characteristics of diffusive transport processes in plasmodesmata can be derived from physical theory of nanoscale pores with charged walls (Fig. 1). In our initial tests of the predictions, we visualized symplasmic movements of differently sized and charged molecules in a tissue known to undergo a strong reduction of plasmodesmal size exclusion limits during maturation. First, we introduced a gene encoding the fluorescent protein mVenus into epidermal pavement cells of sink and source regions of *Nicotiana benthamiana* leaves by *Agrobacterium*-mediated transient transformation. In sink regions, a halo of outwardly decreasing fluorescence intensity developed around individual cells that produced mVenus, indicating that the protein diffused into neighboring cells (Fig. 2*A*). The comparatively strong signal originating from nuclei facilitated the distinction of fluorophore-specific fluorescence from unspecific autofluorescence of cell walls and cuticle. In contrast, the fluorescence signal remained restricted to individual cells in source regions (Fig. 2*A*). Since mVenus has the same mass as GFP (27 kDa), these results confirmed that plasmodesmal size exclusion limits

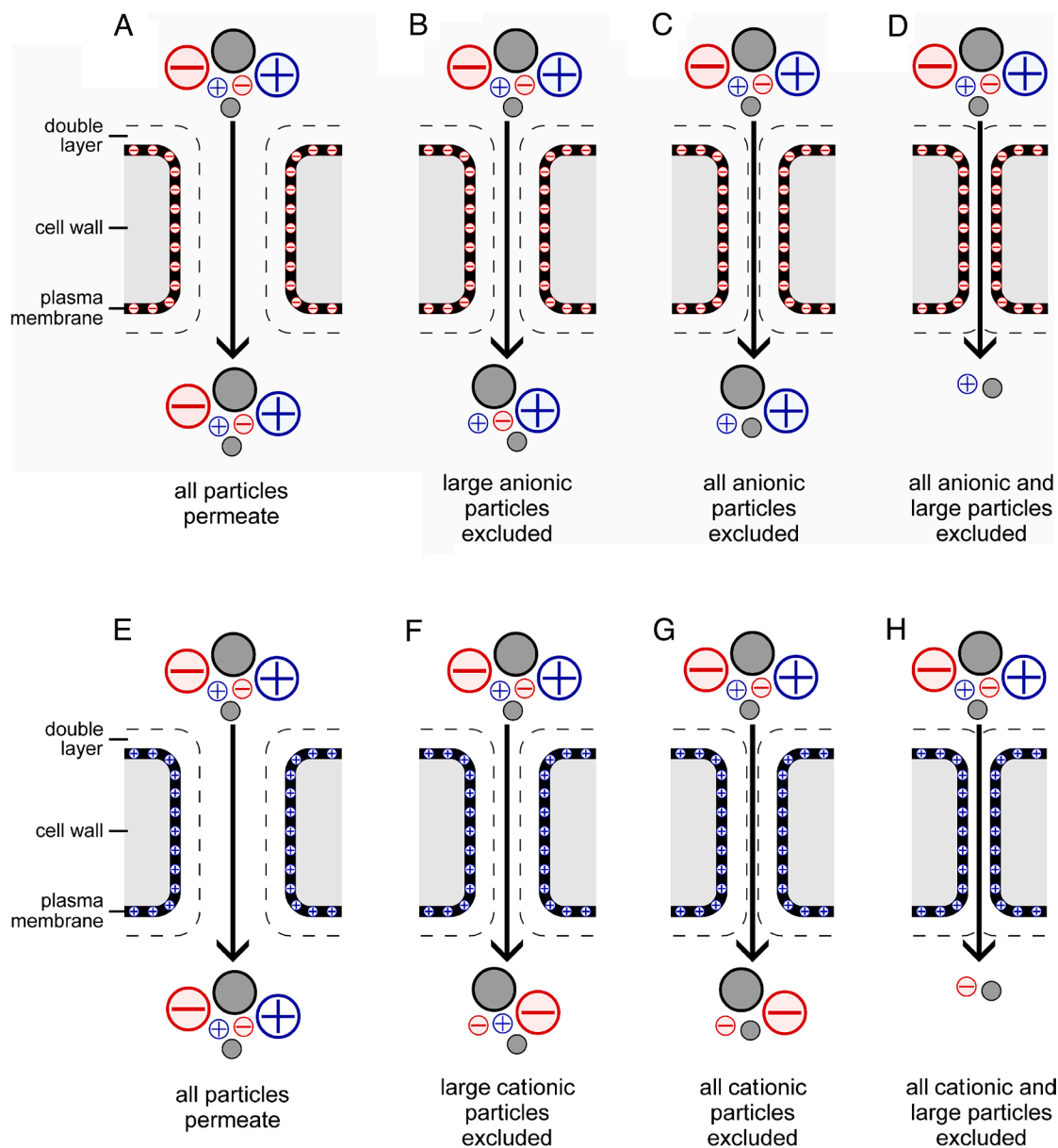


Fig. 1. Electrostatic effects in nanoscopic, membrane-lined pores. Schematic sections of cell wall pores lined with an anionic surface (e.g., plasma membrane); pore diameters decrease from (A–D). Large and small cationic, anionic, and neutral solute molecules are also shown. An electrolyte solution in contact with the negative net charge of the membrane generates an electrical double layer that repels anionic solutes and attracts cationic ones. For a given charge density on the membrane and ionic strength of the cytosol, the thickness of the double layer will be the same regardless of the diameter of the pore. Therefore, the ratio of double layer thickness to pore diameter increases from (A–D). This electrostatic effect will result in larger size exclusion limits for cations and neutral particles than for anions. Consequently, pores in which the thickness of the double layer is large relative to the diameter of the pore are permselective (C and D), whereas this is not so in wider pores (A). Real plasmodesmata are structurally more complex than these models due to the presence of the desmotubule, but the same physical principles apply. Schematic sections of cell wall pores lined with a cationic surface; pore diameters decrease from (E–H). Otherwise all principles apply as in A–D, just with opposite charge.

were reduced during the sink–source transition in *Nicotiana* leaves (21). mVenus carries a slightly lower net charge than GFP at pH_{cyt} (−7 compared to −8). To see whether their negative net charge is critical for the behavior of these anionic proteins, we repeated the experiments using (pos36)GFP, a slightly heavier (28 kDa) derivative with an estimated net charge of +36 at pH_{cyt} (41). The results (Fig. 2A and SI Appendix, Fig. S2 for statistics) were similar as with mVenus, indicating that the ability of GFP-sized proteins to pass through plasmodesmata in these cells was not controlled by electrical charges.

The decrease of the size exclusion limit to below the size of proteinaceous fluorophores during the sink–source transition in leaves seems equivalent to the transition from A, E (sink region)

to D, H (source region), respectively, in our simplified model (Fig. 1). To address the predicted permselectivity, we searched for smaller probes carrying opposite charges. We applied positively, net-neutral, and negatively charged amino acids to individual cells and attempted to analyze their movement to downstream cells, but in all cases the amino acids were quickly metabolized before an appropriate detection limit could be reached. The same was true for deuterated natural components in an attempt to track them via Raman confocal microscopy. Despite various attempts, we were unable to establish a method to track natural molecules at the required speed and subcellular resolution, primarily because of rapid metabolic turnover. Therefore, we searched for alternatives possessing the following key features: a) the probe may not be

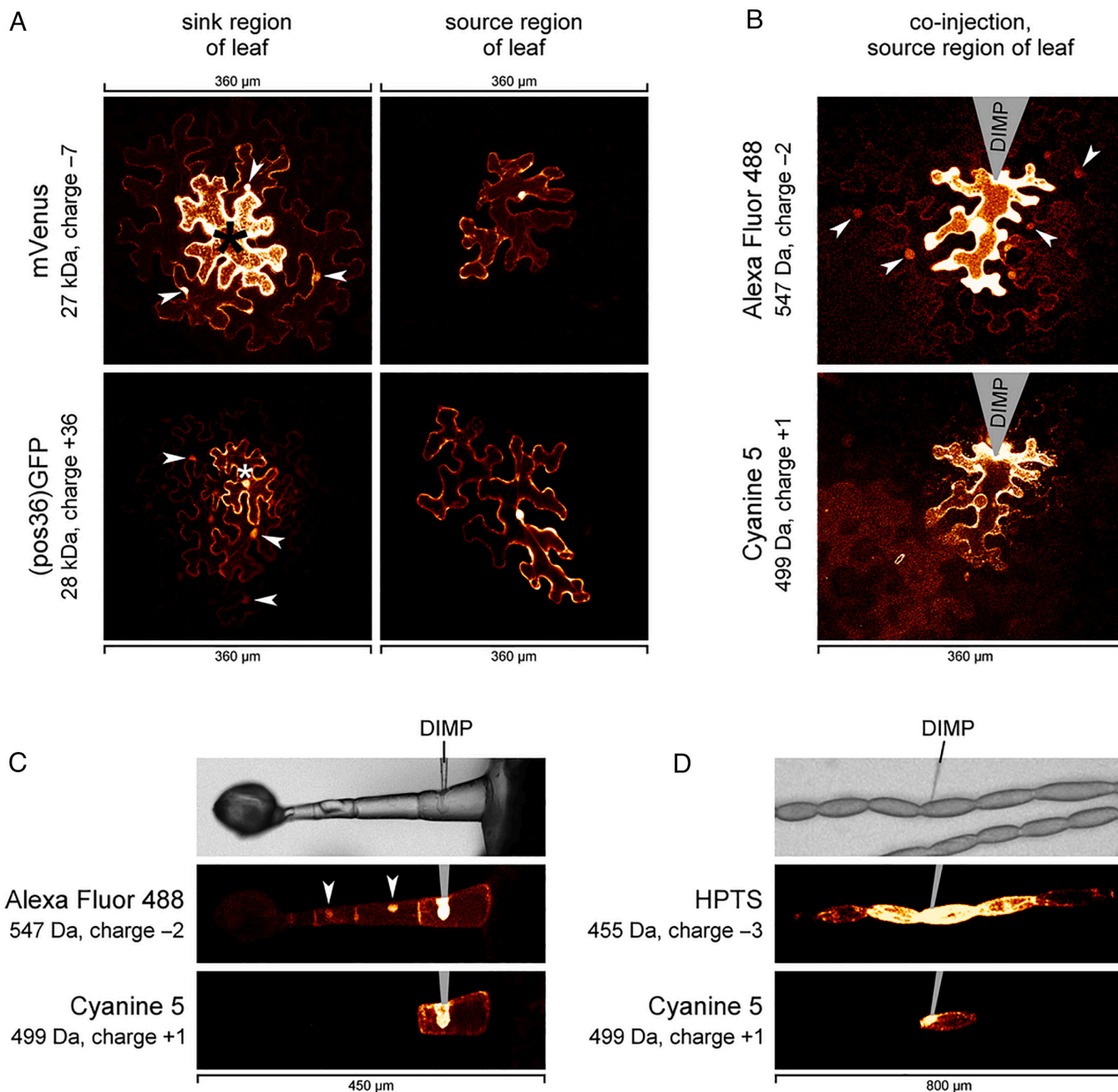


Fig. 2. Qualitative analysis of various fluorophores in three tissue types. (A) Fluorescence signals derived from a negatively charged (Top) and a positively charged (Bottom) GFP derivative in epidermal pavement cells of the sink region (Left) and the source region (Right) in developing *Nicotiana benthamiana* leaves. In all cases, a single cell expresses the gene encoding the fluorescent protein (these cells are marked by asterisks in the micrographs on the Left). With both the anionic and the cationic fluorescent proteins, weak fluorescence becomes visible in the neighboring cells including the nuclei (examples marked by arrows) in the sink region (Left). In contrast, the fluorescent signal remains restricted to a single cell in the source region (Right). (B) Diffusive injection micropipettes (DIMP) are used to simultaneously inject the anionic Alexa Fluor 488 and the cationic Cyanine 5 into pavement cells in the source region of *N. benthamiana* leaves, which results in the movement of the anionic but not the cationic tracer into neighboring cells, as indicated by the appearance of fluorescent nuclei (examples marked by arrows). For the experiments, the tissue is slightly tilted which may result in visible autofluorescence of the cuticle on one side of the sample (note the ledges in the guard cells). As in GFP experiments, fluorescence within nuclei is used to determine diffusion. Fluorescence of the DIMP is masked by overlays. (C) Analogous DIMP experiments in *Nicotiana tabacum* epidermal trichomes produce the same result: the anionic Alexa Fluor 488 but not the cationic Cyanine 5 moves from cell to cell. (D) Analogous experiments using *Tradescantia zebrina* stamen hairs confirm the conclusion. When an anionic fluorophore (in this example, HPTS) is coinjected with a cationic one into cells characterized by small plasmodesmal size exclusion limits, only the anionic tracer moves symplasmically.

metabolized by the cells or break down into smaller detectable fragments, b) the probe needs to be detectable at subcellular resolution and at subsecond time resolution, c) there must be sufficient probe variety so that multiple probe sizes and probe charges may be tested, d) the probe may not be cytotoxic.

The only probes we were able to identify that fulfill all requirements were pure fluorochromes (preparations of fluorochromes

conjugated to molecules such as dextrans may vary in size and are not suitable). We collected information on fluorochromes that have been extensively used in previous studies and whose general properties, especially with regard to cell toxicity, have been determined. We identified a group of 19 fluorochromes that covered all basic requirements and of which molecular information such as charge, pKa(s) hydrodynamic radius and molecular structure,

etc., were readily available (many suitable commercial fluorochromes had to be excluded because necessary information is proprietary). We excluded probes that were known to cross the plasmamembrane such as diacetate derivatives. In addition, we excluded JC-10 because in our hands this dye appeared to move into the apoplast and diffused through the extracellular space underneath the cuticle. To ensure cell health after impalement of cells with microcapillaries, we monitored cyclosis—a generally accepted sign for cell health—for a minimum of 30 min.

Since shifts in intracellular hydrostatic pressure (turgor) affect plasmodesmal transport rates (48), we used diffusive injection micropipettes (DIMPs), a technique we had developed to prevent turgor-induced artifacts [(49, 50); ref. 51 provides a detailed evaluation of potential artifacts generated by the introduction of fluorophores through microdiffusion-pipettes and transient expression by *Agrobacterium*]. We coinjected the anionic Alexa Fluor 488 and the slightly smaller, cationic Cyanine 5 into pavement cells of source tissues. The negatively charged fluorophore moved to neighboring cells, whereas the positively charged one remained restricted to the injected cell (Fig. 2B). To exclude that this observation reflected a peculiarity of pavement cell plasmodesmata, we repeated the experiment in *N. tabacum* trichomes, and obtained the same result (Fig. 2C). Next, we performed analogous tests using *Tradescantia* stamen hair cells, a classical model system for plasmodesma studies with reported plasmodesmal size exclusion limits of under 12% of the mass of GFP (27–29). Again, only the anionic but not the cationic fluorophore moved (Fig. 2D).

Hypothetically, the failure of cationic fluorophores to move through plasmodesmata could be due to their binding to anionic membrane surfaces, which, if it occurred within plasmodesmata, could obstruct the cytosolic sleeve. However, such obstruction should also block the movement of coinjected, similarly sized anionic fluorophores, which we never observed independent of the combination of fluorochromes (Fig. 2B–D).

We conclude that narrow plasmodesmata (i.e., plasmodesmata with size exclusion limits significantly below the molecular mass of GFP) are permselective, as predicted by physical theory of electrostatic effects in nanopores (Fig. 1). Unexpectedly, though, the polarity is reversed: Rather than being selective for cations as current models of plasmodesma structure imply (Fig. 1A–D), narrow plasmodesmata are anion-selective as depicted in Fig. 1E–H. Our equivalent observations in three distinct cell types (Fig. 2B–D) suggest that this anion-selectivity is not specific to certain tissue types but rather occurred by physical necessity. We decided to analyze plasmodesmal permselectivity quantitatively, testing a larger variety of small fluorophores in the stamen hairs of *Tradescantia zebrina*.

Model System for Quantitative Studies: Structure of *Tradescantia* Stamen Hairs.

As a basis for quantitative analyses, we first established pertinent structural features of our experimental material. Stamen hairs used for experimentation were 2.5 to 4 mm long and consisted of 12 to 20 serially arranged cells (Fig. 3A). Cells were constricted toward the cross-walls between them (arrows in

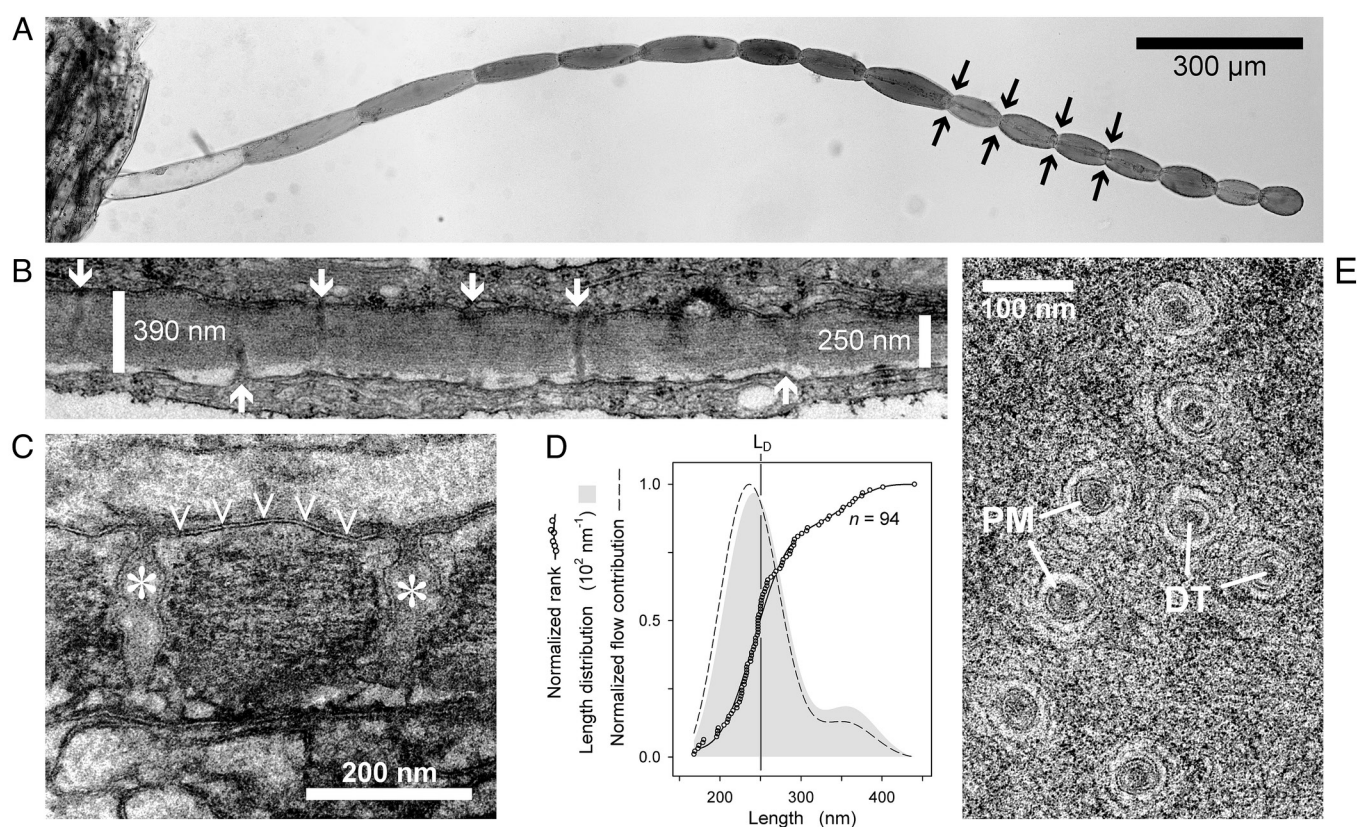


Fig. 3. Structure of *Tradescantia* stamen hairs and the cell walls between their cells. (A) Live stamen hair consisting of 16 serially arranged cells (light micrograph). Cross-walls between cells are located at conspicuous constrictions of the cell file (examples marked by arrows). The hair still is connected to the stamen (Left). (B) Electron micrograph of a cross-sectioned cell wall between two cells, showing numerous plasmodesmata (arrows) and variable wall thickness. (C) Centrally sectioned plasmodesmata reveal the continuity of the plasma membrane (double line; arrowheads) into the widening and irregularly shaped plasmodesmal channels (asterisks). (D) Dependence of plasmodesma length on wall thickness presented as the length distribution (gray shade; derived from the normalized ranks of 94 measured plasmodesmata [circles]), and the estimated relative contribution of plasmodesmata of different lengths to diffusional current between cells (dashed line). The same total current would result if all plasmodesmata had the same effective length, L_D , of 250.4 nm. (E) Cross-sections of plasmodesmata show the peripheral plasma membrane (PM) and central desmotubule (DT). Cell-to-cell transport occurs in the cytosolic sleeve between these two membraneous structures.

Fig. 3A); cell diameters at these cross-walls were 23 to 42 μm with a mean of 32 μm (determined on light micrographs of 4 stamen hairs, $n = 60$). In the transmission electron microscope, the cross-walls exhibited variable thickness over short distances (Fig. 3B), and numerous plasmodesmata were visible (Fig. 3B and C). Consequently, the profile of plasmodesma lengths ranged from 170 nm to 440 nm ($n = 94$), with a modal length of about 245 nm (Fig. 3D). Based on this length profile and the dependence of diffusive current through cylindrical pores (see *Methods* for details), a profile of the relative contributions of plasmodesmata of different lengths to total diffusive current past the cross-walls was derived (Fig. 3D). The effective plasmodesma length (i.e., the length that, if it would apply to all plasmodesmata, would result in the same overall current) was 250.4 nm.

In sections more or less parallel with the plane of the transverse walls, plasmodesmata showed the plasma membrane and desmotubule, in which the ER membrane could not be identified unambiguously, and an apparently open cytosolic sleeve between them (Fig. 3E). The apparent width of the open sleeve was 6.2 ± 2.0 nm (mean $\pm$ SD). However, this value refers to sleeve width in the interior part of the plasmodesmal pore. Close to the pore openings, pore width decreased significantly (Fig. 3C), implying narrower sleeves in these regions. We estimated plasmodesma density to be $11.4 \mu\text{m}^{-2}$ ($n = 3$, range $8.4 \mu\text{m}^{-2}$ to $14.4 \mu\text{m}^{-2}$; *SI Appendix*, Fig. S1), in good agreement with an earlier study (52). Thus, a transverse wall of average radius (16 μm ; area $\sim 800 \mu\text{m}^2$) was expected to carry $\sim 9,000$ plasmodesmata.

Quantitative Analysis: Fluorophore Movements in *Tradescantia* Stamen Hairs. To visualize symplasmic transport through plasmodesmata in *Tradescantia* stamen hairs, we introduced 19 membrane-impermeant fluorophores (for a complete list, see Fig. 4 below) that differed in net charge (from -3 to $+2$ at pH 7; fluorophores of net charge 0 were zwitterionic and also membrane-impermeant) and molecular mass (237 to 1,027 Da) into stamen hair cells. The hairs remained connected to the stamen during the experiments, as hair removal affects plasmodesmal conductivity (28). Representative individual datasets are presented in Fig. 4A and B.

We attempted to co-inject pairs of dyes, ideally a negative one combined with one net neutral or positive dye when possible. Based on spectral dye properties (to prevent cross talk causing false positive results) we applied 30 different dye combinations. Only fluorescein and lucifer yellow (in three of seven cases) were injected individually because of their broad emission spectrum (Dataset S1).

Tracers frequently employed for plasmodesma studies such as Lucifer Yellow CH (charge -2 , 443 Da) moved readily from the injected cell into adjacent ones, and so did HPTS (-3 , 455 Da), a fluorophore of superior characteristics (26) that rarely has been used due to the lack of appropriate filters in earlier microscopes (Fig. 4B). The largest anionic fluorophore tested, Alexa Fluor 633 (-2 , $\sim 1,027$ Da), remained restricted to the injected cell (Fig. 4B). None of the applied cationic fluorophores seemed able to leave the injected cell, independently of their molecular masses (237 to 499 Da). To quantify the symplasmic cell-to-cell conductivity for each of the 19 fluorophores, time-courses of fluorescence intensity were determined in the injected cell and a pair of neighboring cells on each side (Fig. 4C). Under the assumptions i) that fluorescence intensity corresponds to fluorophore concentration, ii) that no significant fluorophore concentration gradients exist in the cytoplasm of a single cell, and iii) that diffusive currents of fluorophores occurred unidirectionally from the injected cell to

both the left and right, the kinetics of the fluorescence signals in two series of three cells each (*ic-rn-rnn* and *ic-ln-lnn*; Fig. 4C) allowed the estimation of plasmodesmal permeability, k (Fig. 4D; see *Methods* for details).

Plasmodesmal permeability clearly depended on electrical charge while molecular masses had no apparent effects (Fig. 5A; see Fig. 5B and C and *SI Appendix*, *Statistics* for formal analyses of the correlations between permeability and charge or mass). For example, the ratios of the permeabilities for the almost identically sized (443–455 Da) Rhodamine 6G, Rhodamine B, Lucifer Yellow CH, and HPTS were 1:2:12:43 (that is, HPTS spread from cell to cell over 40 times faster than Rhodamine 6G), corresponding to the charges of these molecules ($+1$, 0 , -2 , -3 ; Fig. 5A). The correlation was not perfect, though, as the monovalently anionic Sulforhodamine 101 (607 Da) did not move at notable rates, in contrast to other anionic dyes of similar sizes (Fig. 5A). As mentioned above, the conductivity for the divalently anionic Alexa Fluor 633 of just over 1 kDa also was low, while the slightly smaller CF 488A (~ 928 Da) moved comparatively freely (Fig. 5A). This suggested that under our experimental conditions, plasmodesmata had a size exclusion limit for anionic particles of about 1 kDa, which was in accord with previous studies. Evidently, a much lower size exclusion limit of under 300 Da applied to cationic fluorophores, as even the smallest ones tested, the cationic Acridine Yellow and Safranin O (both $+2$) of roughly one third the mass of CF 488A, did not travel symplasmically at noticeable rates (Fig. 5A).

Taken together, our quantitative analysis corroborated the conclusion that narrow plasmodesmata are permselective, but with the opposite of the expected polarity (compare Fig. 1D). The conundrum would be solved, of course, if membrane surfaces within plasmodesmata were positively rather than negatively charged (compare Fig. 1H). Assuming a concentric plasmodesmal pore structure with cationic membrane surfaces, 3 nm cytosolic sleeve width with an electrical double layer thickness of 1 nm, and a hydrodynamic radius of 1.5 nm for the largest tested fluorophore (corresponding to the steric size exclusion limit of 3 nm; Fig. 5D), we modified previously developed theory (53) to calculate expected permeabilities k as a function of both particle size and charge (for details, see *Methods*). For the 19 fluorophores tested, expected and measured k values matched reasonably well (Fig. 5D), demonstrating that the assumption of positively charged surfaces within plasmodesmal pores provides a physically sound explanation for the observed anion-selectivity of narrow plasmodesmata. We emphasize that these cationic surfaces do not necessarily have to be membrane surfaces; proteins subdividing a wider cytosolic sleeve into narrower transport channels could serve the same function.

Discussion

In living cells, large proportions of the organic and practically all inorganic components of the cytosol are electrically charged, and so are the surfaces of biological membranes. The implications for cell-to-cell transport via membrane-coated nano-scale channels have not always been appreciated. In the past, the experimental recognition and thus the awareness of electrostatic effects in plasmodesmata might have been obfuscated by the large variability of plasmodesmal structure between cell types and developmental stages (8). Physical principles imply that comparatively small changes of pore size will result in qualitative modifications of the conductivities for various particles. In

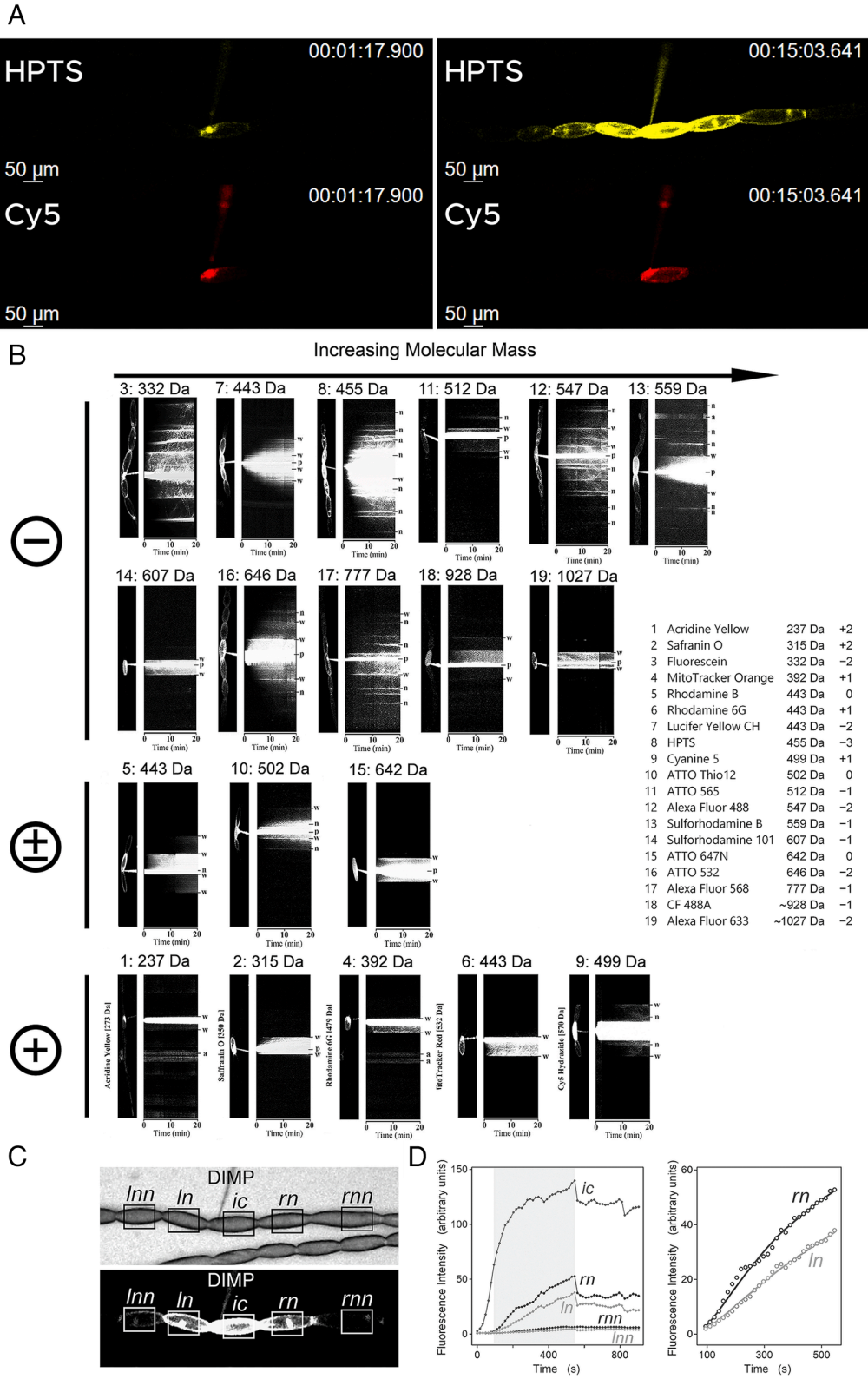


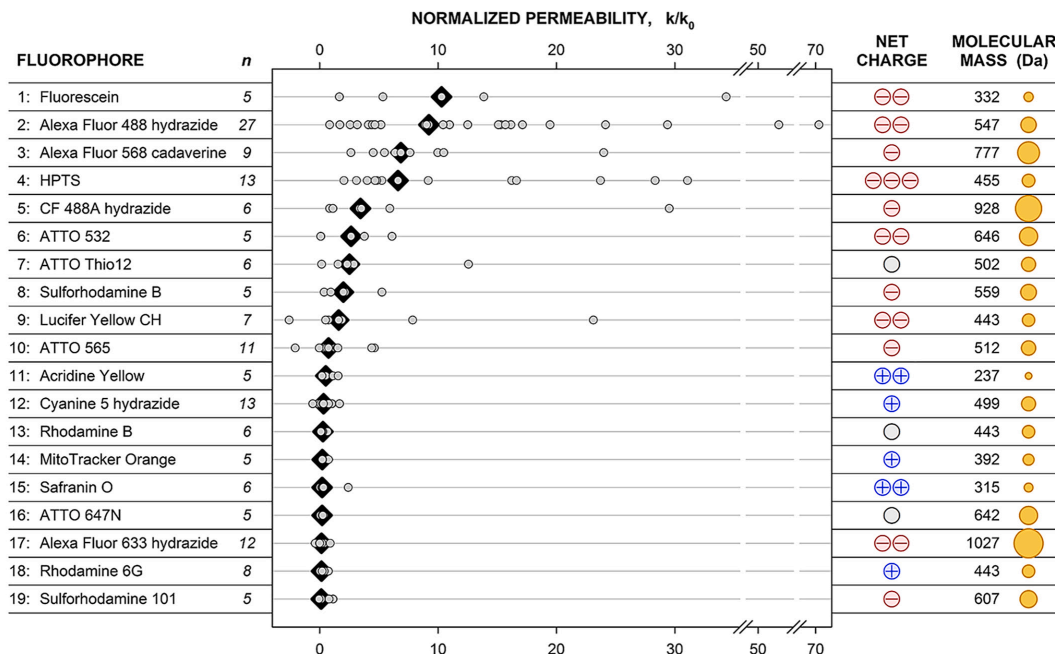
Fig. 4. Quantitative analysis of fluorophore movement in *Tradescantia* stamen hairs. (A) Two images extracted from [Movie S1](#) showing codelivery of the negatively charged HPTS which moves to neighboring cells and the positively charged CY5 which does not move. (B) Tests with fluorophores that are anionic (−), net neutral (±), or positively charged (+) at pH 7; molecular mass increasing from left to right. Each panel comprises a fluorescence image 20 min after DIMP impalement on the *Left* (the DIMP is impaling a cell from the *Right*), and a rotated time-lapse stack (78 images) visualizing the dispersal of fluorescence over 20 min after impalement on the *Right*. Most small anionic fluorophores diffuse along the cell file while the largest one tested (Alexa Fluor 633; ~1,027 Da) remains within the impaled cell. In analogous tests with fluorophores that are net neutral or positively charged at pH 7 significantly less movement is observable. (C) Live stamen hair (*Top*, brightfield micrograph; *Bottom*, fluorescence micrograph). A fluorophore is injected into the cell marked *ic* by a DIMP. Its movement into the left and right neighboring cells (*In* and *Inn*, and *rn* and *rnn*, respectively) is monitored by quantifying fluorescence intensity in the marked rectangular fields. (D) Time-courses of fluorescence intensity in the five cells (*Left* graph). A model of diffusive fluorophore translocation was fitted to the data from all five cells for the period from the occurrence of fluorescence in cells *In* and *rn* to an artificial shift in fluorescence intensity due to refocusing the microscope (gray in *left* graph). The model output for cells *In* and *rn* (fitted lines on the *Right*) provided an estimate of symplasmic permeability, *k*.

general, electrostatic effects of any fixed charges in the pore walls (i.e., the cytosol-exposed membrane surfaces in plasmodesmata) are negligible in pores of radii significantly wider than the electrical double layer covering the pore wall's surface (Fig. 14). Narrow pores in which the electrical double layers of opposing walls overlap will be penetrated preferentially by counterions and uncharged particles of sufficiently small size, while coions

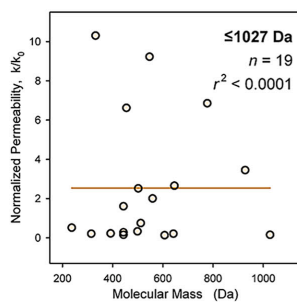
are excluded regardless of size (Fig. 1 C, D, G, and H). In intermediately sized pores, size-exclusion limits for counterions and coions may differ as they are determined sterically and electrostatically, respectively (Fig. 1 B and F).

Despite the unexpected insight that the surfaces bordering the open pathway for diffusive transport in plasmodesmata must be positively rather than negatively charged, our experimental results

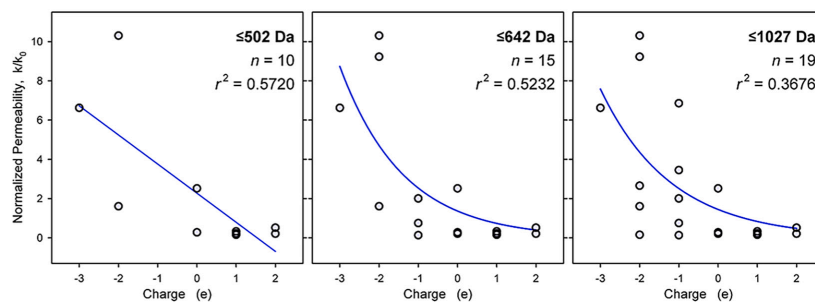
A



B



C



D

Theoretical Model:

plasma membrane with +40 mV surface potential

1 nm double layer

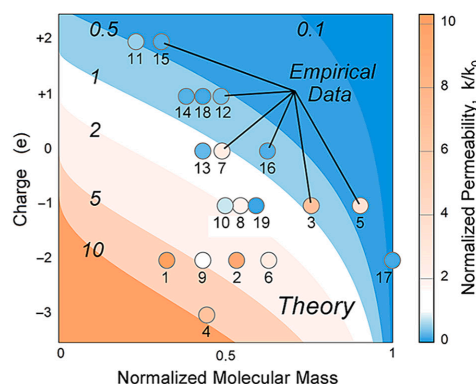
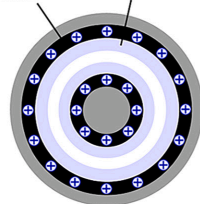


Fig. 5. Quantification of plasmodesmal permeability in *Tradescantia* stamen hairs. (A) Summary of experimental results. The 19 fluorophores and their numbers of biological replicates (n) are listed on the *Left*. The plot in the center presents the normalized plasmodesmal permeability (k/k_0) for each fluorophore, showing measured values for each biological replicate as small gray circles and their median as a black diamond. Note that fluorophores are arranged according to their corresponding permeabilities, with the fastest moving ones on *Top*. Electrical charge and molecular mass of each fluorophore are visualized on the *Right*. Negative charge dominates among fast moving fluorophores, whereas there is no apparent correlation of permeability and fluorophore mass. (B) Normalized plasmodesmal permeability k/k_0 is not correlated with the molecular masses of the 19 fluorophores. The straight line indicates the constant number with minimum sum of residual squares (0th order polynomial regression), the best fitting function according to AIC_C comparison. For further details, see *SI Appendix, Statistics*. (C) Normalized plasmodesmal permeability correlates with the electrical charges of the fluorophores. Blue lines show the best fitting regression functions according to the AIC_C model choice criterion. The three graphs represent subsamples of the fluorophores defined by maximum molecular mass, ranging from 502 Da (*Left*) to 1,027 Da (*Right*). The coefficient of determination, r^2 , decreases with increasing molecular mass limit (for details, see *SI Appendix, Statistics*). (D) A model plasmodesma, in which the cytosolic sleeve of 3 nm width is bordered by positively charged membranes and electrical double layers of 1 nm thickness, is shown on the *Left* (for details, see *Supplementary Information* section "Comparison of observed plasmodesmal permeabilities with theory"). Normalized theoretical plasmodesmal permeabilities (k/k_0) derived from this model in dependence of electrical charge and normalized molecular mass of transported particles are indicated by the background coloration in the plot on the *Right* (isolines mark values of 10, 5, 2, 1, 0.5, and 0.1, as indicated). The 19 fluorophores, identified by numbers as in (A), are shown as circles with colors corresponding to the normalized measured permeabilities according to the scale on the *Right*. Similarity of a circle's color with the background indicates correspondence of expected and measured values.

generally are in line with predictions derived from theory. We caution, though, that the overall picture is complicated by poorly understood additional effects, as suggested by the size-independent variation of the permeabilities for fluorophores of the same charge (for examples, see the mono- and divalent anions in Fig. 5A). Similarly, in experiments with artificial “ultra-nanoscale” (<5 nm) channels, charged and uncharged organic molecules exhibited behaviors that were not predicted by current theory (54). Neither molecular weight nor hydrodynamic radius or electrical net charge alone define a molecule’s movement abilities in solution and in pores, because the three parameters plus molecular 3D structure, charge exposure, etc. all affect the ways in which molecules interact with each other. Therefore, it is hardly surprising that the ability to traverse plasmodesmata does not seem to depend on net charge alone in the fluorophores we tested. This further highlights that the notion that plasmodesma functionality could be quantitatively characterized by a single, geometrically defined size-exclusion limit is overly simplistic and should be abandoned.

The implications are significant for many fields. Plasmodesmata with small size exclusion limits are common in a large fraction of tissues. For example, the path from photosynthesizing mesophyll cells to the sieve element/companion cell complex in plants that load the phloem by the polymer trap mechanism (55) requires that sucrose (342 Da, hydrodynamic diameter ~0.42 nm) can diffuse through plasmodesmata while stachyose (666 Da, hydrodynamic diameter ~0.6 nm) (56) cannot. In the same tissue, long distance signals are thought to communicate nutrient demands via peptides such as C-terminally encoded peptides (CEPs). Those are transported from roots to shoots via the xylem to communicate nitrate starvation (57). To induce a translation of the signal, the peptides have to reach living cells. The charge of CEP1, for example, is +1 (58). The positive charges of this peptide suggests that passive diffusion through plasmodesmata may be unlikely and transporters may be required. The transcription factor Elongated Hypocotyl 5 (HY5), on the other hand, is transported from shoots to roots via the phloem to regulate the expression of nitrate transporters (57). Plasmodesmata in the root unloading zone are very large (23, 59) and despite the charge of +6 of the 18 kDa HY5 protein, it is very likely to diffuse unhindered in the unloading tissue.

Similarly, current models explaining the generation of concentration gradients across organs of charged phytohormone molecules such as auxins (43) will have to take plasmodesmal charge selectivity into account. In live tissues, auxins like indole-3-acetic acid (IAA) are mostly conjugated to a variety of molecule types (60), and conjugation may affect plasmodesmal permeability for auxins due to not only increased molecule size but also changes in electrical charge. Such regulation by charged conjugates of a compound’s ability to permeate plasmodesmata may occur in cytosolic molecules more generally.

The experimentally observed plasmodesmal selectivity for anionic fluorophores (Figs. 2 B–D, 4 A and B, and 5 A–C) could be explained by positively rather than negatively charged membrane surfaces within plasmodesmata. Our analysis based on electrokinetic theory is consistent with this interpretation (Fig. 5D). However, the proportion of negatively charged phospholipids in the plasma membrane appears to be similar within and outside of plasmodesmata (61). Therefore, we hypothesize that the positive charges on intraplasmodesmal structures demanded by our results may be localized on proteins rather than lipid molecules.

Plasmodesmata are much more complex than the simple annular channels depicted in simplified models such as our Figs. 1 and 5D. According to current models, the desmotubule is tethered to the plasma membrane by proteinaceous spokes (4–6, 62); plasmodesmata represent a specialized type of plasma membrane/ER contact site (63). The tethering of the ER to the plasma membrane at interorganelle contact sites is mediated by several protein families, including proteins carrying C2 domains that bind to anionic membrane lipids through Ca^{2+} -dependent or Ca^{2+} -independent mechanisms (64, 65). In the last decade, multiple C2 domains and transmembrane region proteins (MCTPs) have been revealed to form the spoke-like tethers in plasmodesmata that control plasma membrane–desmotubule spacing (10, 66), while the related synaptotagmin family with single C2 domain regulates plasmodesmal permeability by helping to modulate aperture size (67).

Recent studies revealed additional details of plasmodesmal structural organization and interactions of the ER–plasma membrane tethering MCTP family. AlphaFold-based modeling and coarse-grained simulations of *Arabidopsis* MCTP4 show a three lobed C2 architecture facing the cytosol, and an ER-embedded transmembrane region with flexible helix packing yet stable membrane insertion (68). Complementary Martini 3 simulations of the full MCTP4 cytosolic region against plasma membrane-like bilayers reveal multiple docking modes in which one, two, or all three C2 domains can engage the membrane and potentially create a tunable “valve” that adjusts the plasma membrane–ER gap and thus the properties of plasmodesmal transport channels on short time scales (69).

In *Arabidopsis*, common MCTPs thought to be present in plasmodesmata carry net positive charges of +19 (MCTP1/FTIP1), +16 (MCTP3), +16 (MCTP4), +15 (MCTP6), and +24 (MCTP9) at pH 7.2. Fig. 6A shows the exposed surfaces of these proteins along with the related SYT1 single C2 domain-containing protein. In their most compacted form, a highly positively charged surface is clearly present in all MCTPs. We speculate that the electrostatic forces responsible for the observed plasmodesmal selectivity for anionic particles arise from an at least partial shielding of the negatively charged membrane surface by MCTPs (Fig. 6). Fig. 6B is based on the model by Pérez-Sancho et al. (69). The dimensions of desmotubule, cytosolic sleeve, and plasmodesma diameters were measured on electron micrographs (Fig. 3E and Dataset S2). The MCTP4 reconstructions showing surface electrostatics that are integrated in the model are sized to scale. Interestingly, the protein size (a single C2 domain has dimensions of $\sim 2.7 \times 3.2 \times 5.7$ nm based on the smallest C2 in MCTP3) very well spans the cytosolic sleeve (gap space between plasma membrane and desmotubule) and leaves spaces that have at least partially positively charged walls. The resulting pores are much more complex than a straight cylinder or annulus and constitute multiple, individual, or interconnected, unevenly shaped transport channels. Nonetheless, some negatively charged membrane surfaces still face those transport channels. Therefore, we assume that additional positively charged components (which may include other MCTPs or related proteins) are involved in the formation of anion-selective transport channels. Fig. 6C provides a simple representation of our working hypothesis of how positively charged transport channels within the cytosolic sleeve may affect plasmodesma function. Several additional proteins have been localized to plasmodesmata without identification of their exact positions, and we refrain from including more details as it would be overly speculative at this point.

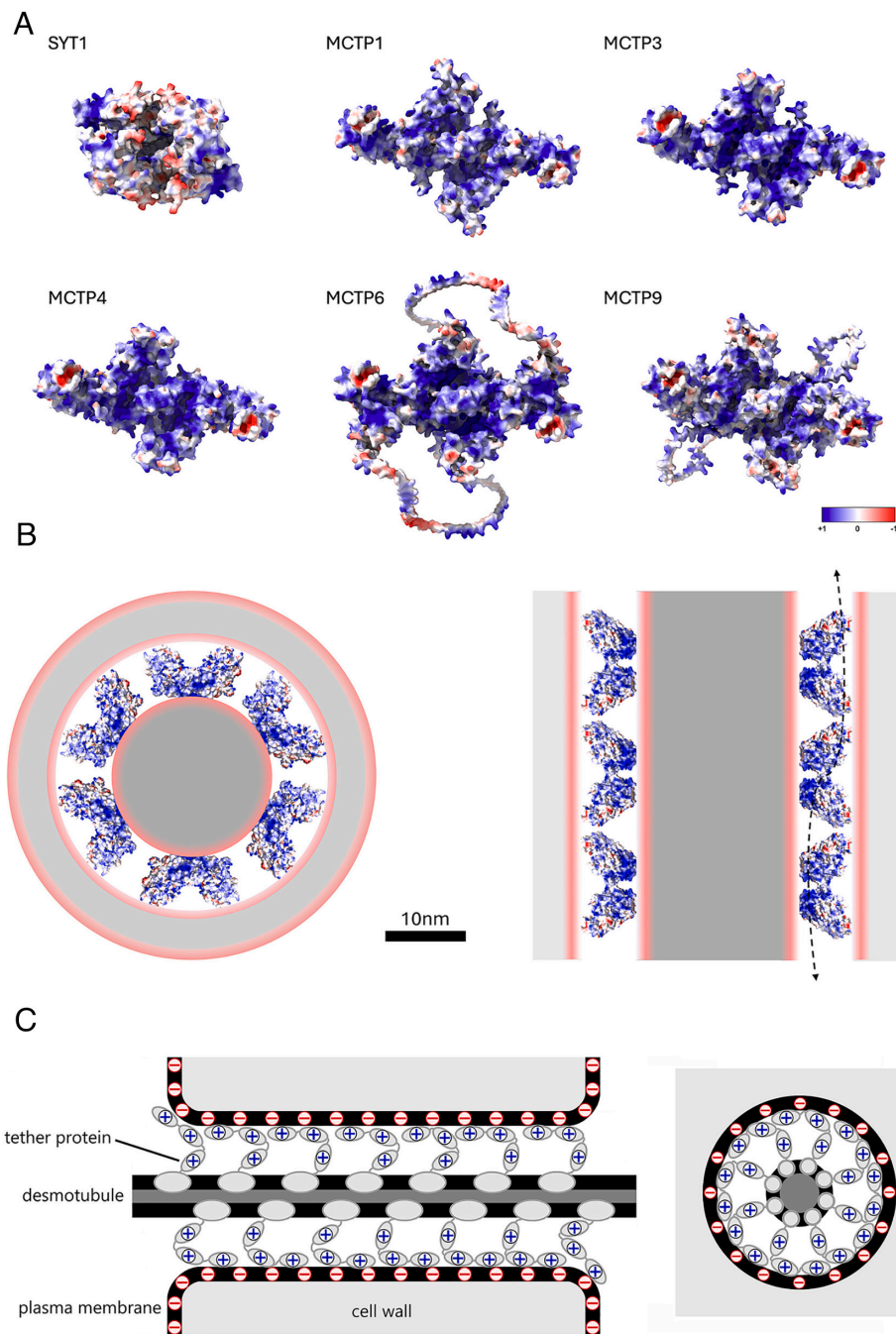


Fig. 6. Plasmodesma model considering electrostatic surface charges. (A) Visualization of surface electrostatics for C2 family ER/PM tether proteins. Molecular surfaces for AlphaFold3 in silico models of homodimers of SYT1, MCTP1, MCTP3, MCTP4, MCTP6, and MCTP9 displayed according to Coulombic electrostatic potential. View for all proteins is perpendicular to the lipid bilayer to show top exposed surface. Color key is the same for all proteins. (B) Plasmodesma model after Perez Sancho et al. (2025) at dimensions as determined by micrograph measurements and protein modeling (69). MCTP4 in its most compacted form spans the gap space and adds significant positive charges to the cytosolic sleeve space. Nevertheless, negatively charged membrane surfaces still face some of the potential transport channel surface indicating that different protein configurations of MCTP4 or yet to be determined components may be present in addition to MCTP4. (C) Tether proteins known to interact with anionic membrane components may provide the positive charges along the cytosolic sleeve that would explain our results. Based on a published structural model (10), a schematic longitudinal section (Left) and a cross-section (Right) of a plasmodesma shows how cationic tethers could balance the negative charges of the plasma membrane and create geometrically complex transport channels within the cytosolic sleeve.

Methods

Quantification of Plasmodesmal Permeabilities in *Tradescantia*. Entire stamens were excised from *T. zebrina* flowers and placed on 2% w/v agar plates so that several stamen hairs lay flat on the agar surface. Agar plates were transferred to the Leica TCS SP8 confocal microscope for fluorophore injection. DIMP were filled with one or, in most experiments, two fluorophores (0.2 mM each, in 100 mM KCl), and were mounted on the MPC200 micromanipulator for single-cell injection. Dye combinations were chosen from 19 fluorophores (Table 1), based on their nonoverlapping excitation and emission spectra. Successful injections into the cytoplasm were defined by the observation of fluorescence in the constantly moving cytoplasmic strands that traverse a cell's central vacuole. Symplasmic movements of fluorophores were documented by time-lapse photography (minimum one image per 15.6 s) with excitation and emission settings that varied based on the fluorophores used. To visualize the dispersal of the fluorescence signals as seen in Fig. 4A and B, time series of fluorescence micrographs were stacked and rotated by 90° using ImageJ (<https://imagej.nih.gov/ij/>).

Total fluorescence intensities were measured in injected cells as well as in the two neighboring cells on each side (for cell identification, see Fig. 4C) in the images of the time-lapse image series described above, using the Leica LAS X software (Leica Microsystems, Wetzlar, Germany). Periods for quantitative analysis were selected in which the fluorescence intensity in the impaled cell had not reached a plateau yet, and in which no artificial shifts of fluorescence intensity occurred due to cells shifting out of focus (Fig. 4D).

Assuming uniform fluorophore concentrations in the cytoplasm of each cell, unidirectional net current from the impaled cell *ic* to the left (neighboring cells *ln*, *lnn*) and right (*rn*, *rnn*), and taking fluorescence intensity as a proxy for fluorophore concentration (*c*), the symplasmic permeability *k_m* of the cell walls separating cell *rn* from its neighbors *ic* and *rnn* can be determined if time courses of fluorescence intensity, *c*(*t*), are known for the three cells. According to a Fick's law-like model of the concentration distribution,

$$\frac{dc_{rn}}{dt} = k_m (c_{ic}(t) - 2c_{rn}(t) + c_{rnn}(t)), \quad [1]$$

Table 1. Small fluorophores tested in this study

Fluorophore	Mass (Da)	Net Charge (at pH 7)	Supplier	Biological Replicates
Acridine Yellow	237	+2	Thermo Scientific	5
Safranin O	315	+2	Sigma-Aldrich	6
Fluorescein (Uranin)	332	−2	Sigma-Aldrich	5
MitoTracker Orange	392	+1	Invitrogen (Molecular Probes)	5
Rhodamine B	443	0	Sigma-Aldrich	6
Rhodamine 6G	443	+1	Sigma-Aldrich	8
Lucifer Yellow CH	443	−2	Sigma-Aldrich	7
HPTS	455	−3	Sigma-Aldrich	13
Cyanine 5 hydrazide	499	+1	Kerafast	13
ATTO Thio12	502	0	ATTO-Tec	6
ATTO 565	512	−1	ATTO-Tec	11
Alexa Fluor 488 hydrazide	547	−2	Invitrogen (Thermo Fisher Scientific)	27
Sulforhodamine B	559	−1	Biotium	5
Sulforhodamine 101	607	−1	MedChemExpress	5
ATTO 647 N	642	0	ATTO-Tec	5
ATTO 532	646	−2	ATTO-Tec	5
Alexa Fluor 568 cadaverine	777	−1	Molecular Probes (Life Technologies)	9
CF 488A hydrazide	~928	−1	Sigma-Aldrich	6
Alexa Fluor 633 hydrazide	~1,027	−2	Invitrogen (Thermo Fisher Scientific)	12

Molecular masses and charges at pH 7 based on the suppliers' information (which sometimes is incomplete) and the PubChem database of the NIH, USA (<https://pubchem.ncbi.nlm.nih.gov/>) are shown with the number of independent biological replicates for each fluorophore.

where changes of fluorophore concentration in cell m over time are considered proportional to the difference in concentrations between its neighboring cells, with the proportionality factor being k_m . The fluorophore concentration in cell m is found by integrating Eq. 1:

$$c_m(t) = c_m(t_*)e^{-2k_mt(t-t_*)} + k_me^{-2k_mt} \int_{t_*}^t e^{2k_mt'} (c_{ic}(t') + c_{mn}(t')) dt'. \quad [2]$$

The permeability k_m was determined by fitting Eq. 2 to the observed time-course of fluorescence intensity in cell m , using the experimentally determined time-courses in cells ic and mn as input parameters, and k_m as the only variable fitting parameter. In the process, the integral in Eq. 2 was evaluated using the *trapz* function of MATLAB (v. R2020a, MathWorks Inc., Kista, Sweden). Fitting was performed using MATLAB *fminsearch* by minimizing the error function:

$$Err_m = \frac{\sum_i (c_m^{data}(t_i) - c_m^{model}(t_i))^2}{\sum_i (c_m^{data}(t_i))^2}, \quad [3]$$

where c_m^{data} and c_m^{model} are the experimentally measured and the modeled concentration, respectively, and t_i refers to the period that was analyzed. The initial estimate of k_m required for this procedure in MATLAB was established by integrating Eq. 1 and solving for k_m with experimental concentration data from the median of

$$k_m = \frac{c_m(t)}{\int_{t_*}^t (c_{ic}(t') - 2c_m(t') + c_{mn}(t')) dt'}. \quad [4]$$

Because this method could be applied to cell ln analogously, we obtained two estimates (technical replicates) of the symplasmic wall permeability in every stamen hair tested, k_m and k_{ln} ; the mean of these two values provided the k estimate from a given stamen hair (biological replicate). Further analyses were based on between 5 and 27 biological replicates per fluorophore. Permeabilities in the range $10^{-5} - 10^{-3} s^{-1}$ are found. These are not inconsistent with estimates of the intrinsic permeability $k = \frac{\sigma ND}{Vl} \sim 10^{-4} s^{-1}$, taking into account the area of the

shared cell wall, $A \approx 800 \mu m^2$, cell wall thickness, $l \approx 250 \text{ nm}$, number of PDs in the shared cell wall, $N \approx 9,000$, cell volume, $V \approx AL$, cell length, $L \approx 200 \mu m$, diffusion constant, $D \approx 10^{-10} m^2/s$, and cross-sectional area of a single PD accessible to the center coordinate of the diffusing particle, $\sigma \approx 10 \text{ nm}^2$.

We assume that effects of cell geometry, such as wall thickness and area, were averaged out.

Statistical Analysis. Because the distribution of the plasmodesmal permeabilities determined in the replicate experiments appeared skewed for most of the fluorophores tested, we used the median permeabilities for all fluorophores, normalized them with respect to the mean permeability for the three neutral fluorophores (k_0), for statistical analyses. Permeabilities of defined groups of fluorophores (e.g., anionic versus cationic) were compared by the nonparametric Mann-Whitney test (online at <http://vassarstats.net>). Various simple polynomials and exponential decay functions were fitted to fluorophore permeability distributions depending on electrical charge or molecular mass using the linear and nonlinear regression routines of SigmaPlot v.11 (Systat Software Inc., San Jose CA, USA), and the resulting fits were compared by AIC_c model selection following Burnham et al. (70). See *SI Appendix, Statistics* for further details.

For details on the generation of transiently transgenic cells, electron microscopy and analysis of plasmodesma structure in *Tradescantia*, comparison of observed plasmodesmal permeabilities with theory, and in Silico Modeling and net charge calculations of C2 family proteins, see *SI Appendix, Methods*.

Data, Materials, and Software Availability. Study data are included in the article and/or [supporting information](#).

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